

PPE COMPLIANCE DETECTION SYSTEM

AI-Powered Real-Time Workplace Safety Monitoring Using Computer Vision and YOLOv8

PROJECT REPORT

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ABSTRACT

Workplace safety is one of the most important aspects of construction sites and industrial environments. Workers are required to wear Personal Protective Equipment (PPE) such as helmets, safety vests, gloves, goggles, and masks to reduce the risk of accidents. However, monitoring every worker manually is difficult, especially in large workplaces, and safety violations can sometimes go unnoticed.

The PPE Compliance Detection System is developed to automate this process using Artificial Intelligence and Computer Vision. The system uses the YOLOv8 model to detect workers and check whether they are wearing the required safety equipment. It can also identify fall incidents and generate alerts whenever a safety violation is detected.

The application supports both image uploads and live camera monitoring, allowing real-time detection of PPE compliance. All detection results and alerts are stored in a database for future reference and analysis. The system is built using FastAPI for the backend, HTML, CSS, and JavaScript for the frontend, Supabase as the database, and OpenCV for image processing.

This project helps improve workplace safety by reducing manual monitoring, providing quick detection of safety violations, and enabling faster response to unsafe situations. It also provides a foundation for future improvements such as supporting multiple cameras and large-scale industrial deployments.

Keywords: *Personal Protective Equipment (PPE), YOLOv8, Computer Vision, Workplace Safety, Object Detection, FastAPI, OpenCV, Supabase, Real-Time Monitoring, Artificial Intelligence (AI).*

PROBLEM STATEMENT

Ensuring that workers consistently wear Personal Protective Equipment (PPE) is a major challenge in construction sites and industrial workplaces. Most organisations still rely on manual supervision to monitor safety compliance, which is time-consuming and prone to human error. Supervisors may miss violations due to large work areas, multiple workers, or continuous operations. This increases the risk of workplace accidents, injuries, and non-compliance with safety regulations. In addition, manual monitoring does not provide real-time alerts or maintain detailed records of safety violations. Therefore, there is a need for an intelligent system that can automatically detect PPE compliance, identify unsafe situations in real time, and assist organisations in maintaining a safer working environment.

PROPOSED SOLUTION

The proposed solution is an AI-powered **PPE Compliance Detection System** that automatically monitors whether workers are wearing the required Personal Protective Equipment (PPE). The system uses the **YOLOv8** object detection model to identify workers and detect safety equipment such as helmets, safety vests, gloves, goggles, and masks in both uploaded images and live camera feeds. Whenever a PPE violation or fall is detected, the system generates an alert and stores the event in a cloud database for future reference and analysis. It also provides a user-friendly dashboard where users can manage cameras, view detection logs, monitor alerts, and access reports. By automating the monitoring process, the system reduces the need for continuous manual supervision, improves the accuracy of safety inspections, and helps organisations maintain a safer and more efficient work environment.

OBJECTIVES

The main objectives of this project are:

- To automatically detect whether workers are wearing the required PPE.
- To identify safety violations and fall incidents in real time.
- To support both image-based and live camera detection.
- To generate alerts whenever a safety violation is detected.
- To store detection records for monitoring and future analysis.
- To reduce manual supervision and improve workplace safety.

SCOPE OF THE PROJECT

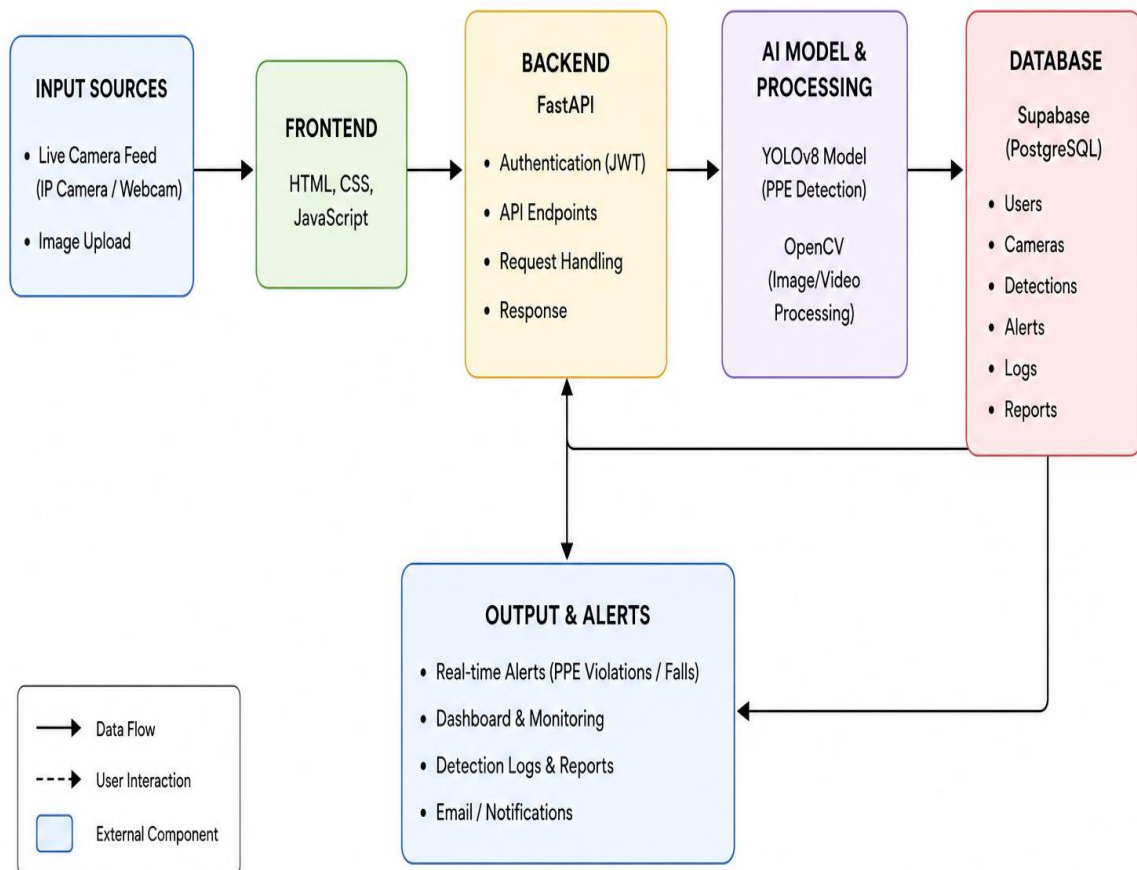
This project focuses on detecting PPE compliance using images and live camera streams. It identifies workers and checks whether they are wearing safety equipment such as helmets, safety vests, gloves, goggles, and masks. The system also detects fall incidents, stores detection results in a cloud database, and provides alerts to improve workplace safety. While the current implementation supports a single live camera at a time, the system is designed so that it can be extended in the future to support multiple cameras and larger industrial environments.

TECHNOLOGY STACK

Component	Technology Used	Purpose
Programming Language	Python	Backend development and AI integration
Frontend	HTML, CSS, JavaScript	Designing the user interface
Backend Framework	FastAPI	Building RESTful APIs and handling backend operations
AI Model	YOLOv8	Detecting PPE equipment and safety violations
Computer Vision	OpenCV	Image and live video processing
Database	Supabase (PostgreSQL)	Storing users, cameras, detections, alerts, and reports
Authentication	JWT (JSON Web Token)	Secure user authentication and authorization
Version Control	Git & GitHub	Source code management and team collaboration
Development Environment	Visual Studio Code	Writing and managing the project code

SYSTEM ARCHITECTURE

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DATABASE DESIGN

The PPE Compliance Detection System uses **Supabase (PostgreSQL)** as its cloud database to securely store user information, camera details, detection events, alerts, reports, and user settings. The database is designed using a relational structure, where each table serves a specific purpose and is connected through unique identifiers. This design ensures efficient data storage, faster retrieval, and reliable management of real-time detection records.

The major tables used in the system are:

Table Name	Purpose
users	Stores user account information and authentication details.
cameras	Stores registered camera information, including camera name, URL, and status.
detection_events	Stores every PPE detection event with the event type and snapshot.
detections	Stores detailed detection results such as detected class, confidence score, and bounding box coordinates.
alerts	Stores generated alerts for PPE violations and fall detection.
user_settings	Stores user preferences such as enabled alerts and violation classes.

FUTURE ENHANCEMENTS

The current PPE Compliance Detection System provides real-time monitoring of safety compliance using AI and computer vision. However, the system can be further enhanced with additional features to improve its scalability, performance, and usability. Some possible future enhancements include:

- 1. Multi-Camera Support:** Enable simultaneous monitoring of multiple cameras from different locations through a single dashboard.
- 2. Advanced Alert System:** Integrate email, SMS, or WhatsApp notifications to immediately inform supervisors about safety violations or fall incidents.
- 3. Mobile Application:** Develop Android and iOS applications for real-time monitoring and instant notifications.
- 4. AI Model Improvement:** Retrain and optimise the YOLO model using larger and more diverse datasets to improve detection accuracy under different lighting and environmental conditions.

CONCLUSION

The **PPE Compliance Detection System** demonstrates how Artificial Intelligence and Computer Vision can be used to improve workplace safety by automatically detecting PPE compliance and fall incidents. By combining **YOLOv8**, **FastAPI**, **OpenCV**, and **Supabase**, the system provides accurate real-time monitoring through both image uploads and live camera feeds. It reduces the need for continuous manual supervision, generates timely alerts, and maintains detection records for future analysis. Overall, the project offers an efficient and practical solution for enhancing safety compliance and can be further extended to support larger industrial environments and advanced monitoring features.